A twelve-variable equation for Con(ZFC)

Quadratic norms, rational Pell integrality,
and a smaller fixed-coefficient compiler

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Abstract

We combine and extend the seven current ProveIt constructions of Diophantine formulations of ZFC consistency. The main result is an explicit fixed-coefficient polynomial in twelve positive integer unknowns, of exact degree 1353, evaluated by 116 binary arithmetic operations. Its absence of positive solutions is provably equivalent in ZFC to Con(ZFC). The previous explicit construction in this compiler family used thirteen positive unknowns. A three-equation presentation has maximum degree 1344 and costs 112 operations. A separately justified signed-domain substitution gives a polynomial in 26 unrestricted integers, of degree 2657 and 168 operations, improving the current 29-integer construction. Independently, a stricter local language and denser synchronization layout reduce the fixed base from $2^{5^{18}}$ to $2^{5^{15}}$: the binary length of $B-1$ is 125 times smaller. All formulas, coefficient recipes, circuit generators, literal transition data, proof notes, and verification scripts accompany this report. The new results are ordinary mathematical proofs, with independent algebraic and finite checks; they are not new end-to-end Lean or Rocq formalizations. We claim improvements to this explicit family, not a global minimum or a proof of consistency.

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1 Results, scope, and comparison

The source snapshot reviewed here is

`f0ae4e0991ca44a7bb4e5671d42e33d775e48035`.

The principal source directory is

`SetTheory/Cardinals/docs/reports/`
`hilbert-tenth-problem/con-zfc-diophantine-forms/`.

At this snapshot it contains packages 09–15, which supersede packages 01–08 [1]. We use the repository's package numbers as source identifiers, without assigning priority beyond the repository's own reconciliation. The comparison concerns actual finite polynomial descriptions, including exact recipes for their fixed integer coefficients.

1.1 What the theorem says

Fix the syntactic proof predicate for ZFC used in the K215 calibration. Then Con(ZFC) means that there is no finite derivation of falsity from those axioms. Our main polynomial P_{12} satisfies

$$\mathrm{ZFC} \vdash \left(\exists \mathbf{x} \in \mathbb{Z}_{>0}^{12}\ P_{12}(\mathbf{x}) = 0\right) \longleftrightarrow \neg\,\mathrm{Con(ZFC)}.$$

Equivalently,

$$\boxed{\mathrm{ZFC} \vdash \left(\forall \mathbf{x} \in \mathbb{Z}_{>0}^{12}\ P_{12}(\mathbf{x}) \neq 0\right) \longleftrightarrow \mathrm{Con(ZFC)}\ .} \tag{1}$$

The four coefficient ports occurring below are fixed integers, not additional quantified unknowns. Strict positivity is part of this statement. A construction over all integers is provided separately in Section 7.

The proof factors through the exact blank run of K215:

$$\text{Diophantine solution} \longleftrightarrow \text{blank K215 halts} \longleftrightarrow \text{a finite ZFC refutation exists.}$$

The second equivalence has existing Lean and Rocq proofs for the literal machine [7]. The first uses the existing inputless compiler and the new mathematical arguments supplied here. We do not infer an internal ZFC derivation merely from a proof assistant's external correctness record: the ordinary arithmetic, finite-computation, and calibration arguments are the arguments to be formalized in ZFC.

1.2 The measured improvements

An operation is one binary $+$, $-$, or $\times$, including multiplication by a fixed numeral. Reuse and the comparisons in a system are free. Degree is total degree in the quantified unknowns. Coefficient bit length is a separate metric.

Construction	Unknowns	Equations	Degree	Ops
Existing package 09 product	13 positive	1	571	84
Existing package 09 lower-degree product	13 positive	1	515	85
New norm system	**12** positive	3	1344	112
New norm product	**12** positive	1	1353	116
New lower-degree norm product	12 positive	1	1177	123
New norm sum of squares	12 positive	1	2688	119
Existing package 14 normalized product	29 integer	1	300	119
New integer product	**26** integer	1	2657	168
New integer system	26 integer	3	2640	164
New integer sum of squares	26 integer	1	5280	171

These variable reductions cost degree and arithmetic operations. The existing 67-operation, 15-positive-variable product remains the best stated operation count in the reviewed family [1, 6]. The new coefficient recipe can also be used with that older product, without changing its degree or operation count.

Coefficient metric	Existing T4	New restricted T4
Local tiles	37,075	5,280
Radix exponent	5^8	5^7
Cell length	5^{10}	5^8
Base B	$2^{5^{18}}$	$2^{5^{15}}$
Bit length of $B-1$	3,814,697,265,625	30,517,578,125

The coefficient descriptions are finite and exact. The final integers are not expanded: even the new $B-1$ needs about 3.81 decimal GB in packed binary. An arithmetic circuit's small description therefore must not be confused with small coefficient height.

1.3 Which existing ideas are combined

Source	Contribution retained	Role in the new result
09	Weighted amalgamation of three Pell roots; weak unit-sign recovery	Starting point for absorbing a fourth coordinate
10, 13	Root elimination and transfer of a coherent coefficient tuple	Separation of arithmetic from the local compiler
11	Head-filtered four-cell T4 language; signed main-root proof	Base for the smaller coefficient language and layout
12	Ordinary strong Pell interface; signed soundness; boundary bootstrap	Soundness of reconstructed roots without assumed signs
14	Seven positive ports, odd multiplier, square power scale	The 26-integer version
15	Explicit low-degree alternatives and careful circuit comparisons	Tradeoff benchmark; no claim to improve its low degrees

The new step is to encode the first root, main root, strong root, and auxiliary root through one supplied integer, while retaining the main root's required congruence. A rational-Pell integrality lemma closes the denominator gap that a bare resultant would leave.

2 The explicit twelve-variable polynomial

Supply exactly the twelve positive integers

$$(J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, r) \in \mathbb{Z}_{>0}^{12}.$$

Fix one complete coefficient tuple

$$(\beta, K_0, M_C, M_F), \qquad \beta = B-1, \quad K_0 = D_C + BD_R, \quad M_F = M_F^{\mathrm{nat}} + B - 1.$$

The new default tuple is defined in Section 8. The original T4 tuple is also valid as a complete alternative. In particular the fourth port is the *shifted* mask.

Define polynomial abbreviations

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & & \\
X &= wq, & Y &= sq^3, & E &= XY, \\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\
H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, & & \\
\ell &= EkY, & A &= 1 + \ell(\ell + k), & G &= 1 + \Delta c^2.
\end{aligned}$$

Set

$$R = \big((q-1)(q-F) + \alpha\big)(q^2-1) + (M_C + qM_F)J, \tag{2}$$

$$N_k = k - hE - R, \tag{3}$$

$$N_t = (K_0 + w)C + q - F - z(q-1). \tag{4}$$

The remaining abbreviations are

$$\begin{aligned}
D &= 1 + \Delta i^2 c^4, & Q &= (\Delta i c^2)^2, & d &= Q - 1, \\
v &= cT, & b &= c + RD, & & \\
e &= Q(Dv^2 + b^2) - 1, & B_{\mathrm{aux}} &= 2Qvb, & & \\
W &= v^2 + D + 2b, & L &= 1 + HW, & S &= X + ac + Hr.
\end{aligned}$$

The compiler base B and the auxiliary cross coefficient B_{aux} are different quantities.

2.1 Quadratic norms as finite polynomial notation

For a polynomial in a formal letter u, reduce it modulo $u^2 - g$ to $p_0 + p_1 u$, and define

$$\mathrm{N}_{g,u}(p) = p_0^2 - g p_1^2.$$

All coefficients may themselves be polynomials. This operation is a finite combination of integer additions and multiplications. It introduces no quantified variable.

With temporary formal letters u, t, put

$$\mathcal{F}(u,t) = \big[d(S - Lu - HLt)^2 - H^2 e\big]^2 - H^4 B_{\mathrm{aux}}^2 D, \tag{5}$$

and define the polynomial

$$N = \mathrm{N}_{G,u}\big(\mathrm{N}_{A,t}(\mathcal{F}(u,t))\big). \tag{6}$$

Thus u, t in (5)–(6) are instructions for constructing a polynomial, not two more unknowns.

The recommended single equation is

$$\boxed{P_{12} = N_k N_t (1 + N) - 1 = 0.} \tag{7}$$

Two useful alternatives are

$$\boxed{N_k = 1, \qquad N_t = 1, \qquad N = 0} \tag{8}$$

and

$$Q_{12} = (N_k - 1)^2 + (N_t - 1)^2 + N^2 = 0. \tag{9}$$

Every exponent in these definitions is fixed. The delivered JSON circuits express them using only binary $+, -, \times$.

2.2 An explicit pair evaluator for the norm

For readers wishing to avoid resultants or formal radicals altogether, work with pairs of integer polynomials and the multiplication

$$(x_0, x_1) \circ (y_0, y_1) = (x_0y_0 + Gx_1y_1,\ x_0y_1 + x_1y_0).$$

Addition, subtraction and scalar multiplication are coordinatewise. This is multiplication modulo $u^2 - G$.

Put $a_0 = H^2L^2A$, $Z = H^4B_{\text{aux}}^2D$, and

$$\begin{aligned}
\mathcal{V} &= (S^2 + L^2G,\ 2SL),\\
\mathcal{C} &= d(\mathcal{V} + (a_0, 0)) - (H^2e, 0),\\
\mathcal{K} &= 4d^2a_0\mathcal{V},\\
(\mathcal{N}_0, \mathcal{N}_1) &= \left(\mathcal{C}^{\circ 2} - \mathcal{K} - (Z, 0)\right)^{\circ 2} - 4Z\mathcal{K}.
\end{aligned}$$

Then

$$N = \mathcal{N}_0^2 - G\mathcal{N}_1^2. \tag{10}$$

This is a closed polynomial recipe in the twelve supplied integers. The identity behind the compact evaluator is

$$(C_0^2 + K_1 - Z)^2 - 4C_0^2K_1 = (C_0^2 - K_1 - Z)^2 - 4K_1Z.$$

It saves six binary gates in the pair implementation. The use of $S + Lu$ in $\mathcal{V}$ instead of $S - Lu$ is harmless because the last norm includes both signs of u.

3 The rational auxiliary-Pell lemma

Lemma 3.1 (Rational Pell ordinate is integral)**.** *Let $m \geq 2$ and V be integers. If $y \in \mathbb{Q}$ satisfies*

$$m^2V^2 - (m^2 - 1)y^2 = 1,$$

then $y \in \mathbb{Z}$.

Proof. Write $y = a_1/b_1$ in lowest terms, with $b_1 > 0$. Clearing denominators shows $b_1^2 \mid m^2 - 1$. Put

$$D_0 = (m^2 - 1)/b_1^2.$$

It is a positive nonsquare integer: if it were a square, then $m^2 - 1$ would be a square. We have two positive Pell solutions

$$m^2 - D_0b_1^2 = 1, \qquad (m|V|)^2 - D_0|a_1|^2 = 1.$$

Neither ordinate is zero. Let $\epsilon = x_1 + y_1\sqrt{D_0} > 1$ be the least positive integral norm-one Pell unit. Write

$$\epsilon^n = \chi_n + \psi_n\sqrt{D_0}.$$

Elementary Pell classification gives indices $j, l \geq 1$ with

$$m = \chi_j, \quad b_1 = \psi_j, \quad m|V| = \chi_l, \quad |a_1| = \psi_l.$$

We need the precise first-coordinate divisibility statement

$$\chi_j \mid \chi_l \quad \Longleftrightarrow \quad l \equiv j \pmod{2j}. \tag{11}$$

Indeed, $\chi_{2j} = 2\chi_j^2 - 1$ and $\psi_{2j} = 2\chi_j\psi_j$ imply $\chi_{n+2j} \equiv -\chi_n \pmod{\chi_j}$. The subtraction identity similarly gives $\chi_{2j-n} \equiv -\chi_n \pmod{\chi_j}$. Reduce l modulo $2j$, then reflect into $0 \le n \le j$. For $n < j$, strict growth gives $0 < \chi_n < \chi_j$, so its signed residue cannot vanish. The only possibility is $n = j$, proving (11).

Thus l is an odd multiple of j. Pell multiplication gives $\psi_j \mid \psi_l$, hence $b_1 \mid a_1$. Coprimality forces $b_1 = 1$. □

The lemma is the specific integrality input needed below. In general, $dy^2 \in \mathbb{Z}$ does not imply $y \in \mathbb{Z}$, and neither does $Hy \in \mathbb{Z}$. The special relation between the two auxiliary Pell coefficients is doing real work.

4 Soundness of the norm reduction

4.1 The inherited arithmetic interface

We state explicitly the interface imported from the source packages. For an authentic coefficient tuple, the normalized blank compiler has positive coordinates $J, F, \alpha, z, i, s, w, \eta, \zeta$ and permits signed h, T, γ, f, τ, y once its signed-domain proof is applied. Its four Pell conditions are

$$\tau^2 - \ell(\ell + k) = 1, \tag{12}$$

$$u^2 - \Delta c^2 = 1, \qquad u = X + ac + \gamma H, \tag{13}$$

$$f^2 - \Delta i^2 c^4 = 1, \tag{14}$$

$$QV^2 - (Q-1)y^2 = 1, \quad V = c(Tf - 1) - Rf^2. \tag{15}$$

Together with $N_k = N_t = 1$, these equations encode blank K215 halting. The same soundness argument allows a common unit sign $N_k = N_t \in \{1, -1\}$ initially, and proves that it is $+1$.

The normalized strong equation maps to the ordinary strong interface by $i_{\text{ordinary}} = \Delta i$. Its coefficient becomes

$$(i_{\text{ordinary}} c^2)^2 = \Delta(f^2 - 1) = Q.$$

This is an integral substitution. We never assume that an arbitrary ordinary multiplier can be divided by Δ.

The relevant full signed proofs are package 12's `ORDINARY_BLANK_PROOF.md` and package 14's `signed_domain_proof.md`; package 09 states the weak-sign version used by its own product [2, 4, 5]. Copies and pinned paths are in the archive. These are substantive inherited mathematical results, not conclusions of the finite tests performed here.

4.2 Bootstrap before reconstructing the roots

Initially suppose $N = 0$ and $N_k = N_t = \varepsilon$, where $\varepsilon \in \{1, -1\}$. This covers the system, and will cover the product after the parity argument in Section 5.2.

The transport factor can be rewritten

$$N_t = (K_0 + w)C + 1 - F - (z-1)(q-1).$$

Since $F, z \ge 1$, if $C \le -1$, then $N_t \le -(K_0 + w) \le -2$, a contradiction. Thus $C \ge 0$. For $N_t = 1$, the stronger equality gives $C > 0$; we retain $C \ge 0$ so the proof also covers the negative provisional sign.

The authentic mask bounds now give

$$q \geq 16, \qquad 0 < R < q^4 - 2, \qquad E \geq q^4 > R + 2.$$

Consequently $A, G, D, Q > 1$, $d > 0$, and $b, v, W, L, H > 0$. These bounds precede the recovery of the main root, so there is no circular assumption about its sign or index.

4.3 One coherent branch of the norm

Over an algebraic closure, the evaluated polynomial N is exactly

$$\prod_{\substack{u=\pm\sqrt{G}\\ t=\pm\sqrt{A}\\ f=\pm\sqrt{D}}} [d(S - Lu - HLt)^2 - H^2 e + H^2 B_{\mathrm{aux}} f]. \tag{16}$$

If it is zero, choose one vanishing factor and put

$$y = (S - Lu - HLt)/H.$$

The chosen roots satisfy

$$u^2 = G, \qquad t^2 = A, \qquad f^2 = D, \tag{17}$$

$$dy^2 = e - B_{\mathrm{aux}} f = Q(vf - b)^2 - 1, \tag{18}$$

$$S = Lu + HLt + Hy. \tag{19}$$

The signs are chosen from one actual zero factor. We do not assume that all independent formal sign choices occur as field automorphisms.

4.4 Uniform bounds for every conjugate

Take a finite normal extension of $\mathbb{Q}$ containing this selected branch. Under every embedding, f goes to a square root of D. For every conjugate y', including a complex one,

$$\begin{aligned} |y'|^2 &\leq \frac{e + B_{\mathrm{aux}}\sqrt{D}}{d} \\ &= \frac{Q(v\sqrt{D} + b)^2 - 1}{Q - 1} < 2(v\sqrt{D} + b)^2. \end{aligned}$$

Since $v^2 + D \geq 2v\sqrt{D}$,

$$|y'| < W = v^2 + D + 2b. \tag{20}$$

In particular,

$$L\sqrt{G} > HW. \tag{21}$$

The other separation estimate is

$$HL\sqrt{A} > L\sqrt{G} + HW. \tag{22}$$

Here is a direct proof. We have $\sqrt{A} > \ell$ and $\sqrt{G} < (a+2)c$, because $c > 1$ and $\Delta = (a+2)^2 - 1$. Also $c < k(Y+1) \leq 2kY$. Therefore

$$\begin{aligned} H\sqrt{A} - \sqrt{G} &> H\ell - (a+2)c \\ &> kY\big(HE - 2(a+2)\big) \\ &\geq kY(H - 2a - 4) = kY(2a - 1) > 1. \end{aligned}$$

Multiplying by $L = 1 + HW > HW$ proves (22). Thus one weight L, used with the factors $1, H$, separates both roots. A second large independent weight is unnecessary.

4.5 Rationality, then integrality

If t were irrational, an automorphism of the normal extension would send t to $-t$. Subtract its image of (19) from the original. The changed t-term has magnitude $2HL\sqrt{A}$, whereas the other terms have total magnitude at most $2L\sqrt{G} + 2HW$, contradicting (22). Thus $t \in \mathbb{Q}$, and $t^2 = A \in \mathbb{Z}$ then implies $t \in \mathbb{Z}$.

If u were irrational, an automorphism would send it to $-u$ and fix the now rational t. The same subtraction would give $2L\sqrt{G} \le 2HW$, contrary to (21). Hence $u \in \mathbb{Z}$. Equation (19) now makes y rational. Since $B_{\mathrm{aux}} > 0$,

$$f = \frac{e - dy^2}{B_{\mathrm{aux}}} \in \mathbb{Q},$$

and $f^2 = D \in \mathbb{Z}$ gives $f \in \mathbb{Z}$.

Now $V = vf - b$ is an integer and $Q = (\Delta ic^2)^2$. Lemma 3.1, applied to (18), proves $y \in \mathbb{Z}$. This final step is the point at which rational recovery alone would otherwise have been insufficient.

4.6 Restore the main congruence

Because $L = 1 + HW$, equation (19) gives

$$u - X - ac = H(r - Wu - Lt - y).$$

Thus

$$\gamma = r - Wu - Lt - y \in \mathbb{Z}$$

restores the main-root coordinate exactly. Furthermore $f^2 = D$ implies

$$vf - b = c(Tf - 1) - Rf^2.$$

The recovered t, u, f, y, γ therefore satisfy all four conditions (12)–(15), with $\tau = t$. Their signs need not be guessed. The signed ordinary interface, after the integral multiplier conversion stated above, applies. It recovers the main root's positive sign, forces the provisional index/transport signs to $+1$, and decodes a blank halting computation. This establishes soundness.

5 Completeness and the single-product finalizer

5.1 Every canonical halting witness gives a new witness

If K215 halts on the blank tape, the inherited inputless compiler produces a canonical positive normalized witness. Write its positive main root as $u = X + ac + \gamma H$ and its first root as τ. Keep the eleven coordinates occurring in the new definitions and set

$$r = \gamma + Wu + L\tau + y. \tag{23}$$

This is positive and satisfies

$$S = Lu + HL\tau + Hy.$$

One factor of (16) is zero. Hence $N = 0$, while $N_k = N_t = 1$ already hold. This proves completeness for the system, the product and the sum of squares.

The witness transformation deletes f, τ, y, γ and adds r: fifteen supplied coordinates become twelve. It makes no claim that the new r is numerically small.

5.2 Why $1 + N$ is a valid unit guard

The pair evaluator writes the inner norm as $\mathcal{N}_0 + \mathcal{N}_1 u$. Its odd coefficient $\mathcal{N}_1$ is divisible by 8 as an integer polynomial. One way to see this is to put $x = S + Lu$. The odd coefficient of x^2 is even, that of $\mathcal{C}^2$ is divisible by 4, and that of $\mathcal{K}$ is divisible by 8. Thus the odd coefficient of

$$(\mathcal{C}^2 + \mathcal{K} - Z)^2 - 4\mathcal{C}^2\mathcal{K}$$

is divisible by 8. This expression equals the optimized pair expression in Section 2. Consequently

$$N = \mathcal{N}_0^2 - G\mathcal{N}_1^2 \equiv \mathcal{N}_0^2 \pmod{64}.$$

In particular $1 + N$ cannot be -1 modulo 4.

At a zero of P_{12}, each of the three integer factors $N_k, N_t, 1 + N$ is a unit. The third must be 1, so $N = 0$ and $N_k = N_t \in \{1, -1\}$. The soundness proof was deliberately established for exactly this provisional sign pattern. It forces both signs positive.

Theorem 5.1. *For either the new complete coefficient tuple or the inherited T4 tuple, the three-equation system* (8)*, the equation $P_{12} = 0$, and the equation $Q_{12} = 0$ have positive integer solutions exactly when blank K215 halts. Their no-solution statements are therefore each provably equivalent in ZFC to* Con(ZFC).

6 Exact degree and circuit size

Every supplied unknown has degree one; fixed coefficients have degree zero. The useful degree ledger is

Expression	c	a, H	Δ	R	ℓ	A, G	Q	D
Degree	5	6	12	4	11	22	46	34
Expression	v	b, W	L	S	a_0	$\mathcal{V}_0$	$\mathcal{V}_1$	$\mathcal{C}_0$
Degree	6	38	44	11	122	110	55	168

The leading term of $\mathcal{C}_0$ comes uniquely from da_0. The even coefficient $\mathcal{N}_0$ has degree 672, with leading term $(\mathcal{C}_0)^4_{\text{top}}$; the odd coefficient has degree at most 605. Thus the term $\mathcal{N}_0^2$ has degree 1344, whereas $G\mathcal{N}_1^2$ has degree at most 1232. There is no top-degree cancellation.

More explicitly, set $q_0 = \beta J$, $k_0 = \eta + \zeta$. Then

$$N_{\text{top}} = 2^{80} i^{48} k_0^{112} w^{112} s^{224} q_0^{832} (q_0 - F)^{16}. \tag{24}$$

It is nonzero for every $\beta > 0$. The leading forms of the outer factors are

$$(N_k)_{\text{top}} = -hwsq_0^4, \qquad (N_t)_{\text{top}} = w(q_0 - F - \alpha) - zq_0.$$

Therefore $\deg P_{12} = 1344 + 7 + 2 = 1353$. For example, the coefficient of

$$J^{853} hz\, i^{48} \eta^{112} w^{113} s^{225}$$

is $2^{80}\beta^{853} \neq 0$. The sum of squares has exact degree 2688.

The literal circuit counts are

Presentation	Multiplications	Additions/subtractions	Total
Three equations	71	41	112
Product polynomial	73	43	116
Sum of squares	74	45	119

The generator checks that every gate, every variable and all four coefficient ports are live. These counts concern the delivered evaluation schedule; no optimal arithmetic-circuit claim is made.

6.1 An optional lower-degree weight

There is a second twelve-positive-variable polynomial that exchanges seven extra operations for lower degree. Keep all other definitions, but replace W by

$$W' = 4T(ic^2) + 2 + 2R(a+6)i^2c^3, \qquad L' = 1 + HW'.$$

The norm construction and product finalizer are unchanged. This gives a product of exact degree 1177 at 123 operations, a three-equation system of maximum degree 1168 at 119 operations, and a sum of squares of degree 2336 at 126 operations.

The shorter degree comes from estimating conjugates relative to $\sqrt{G}$, instead of absolutely. Weak outer soundness and $h > 0$ give $R < c$. With $I = ic^2$, use $\sqrt{G} > ac$, $\sqrt{D} < (a+2)I$, and $\Delta/a = a + 4 + 3/a < a + 6$ to obtain

$$\frac{|y'|}{\sqrt{G}} < \frac{2TI(a+2)}{a} + \frac{2}{a} + \frac{2R}{ac} + \frac{2R\Delta i^2c^3}{a} < W'.$$

Also $H\sqrt{A} > 2\sqrt{G}$, since $E \geq 2$ and $HE \geq 2H > 4(a+2)$. These are sufficient for the same two automorphism contradictions, because

$$H|y'| < L'\sqrt{G}, \qquad HL'\sqrt{A} > 2L'\sqrt{G} > L'\sqrt{G} + H|y'|.$$

The integrality lemma, main congruence and product guard are unchanged. Now $\deg W' = 27$ and $\deg L' = 33$; the norm's leading form is

$$2^{80}i^{48}k_0^{96}w^{96}s^{192}q_0^{720}(q_0 - F)^{16}.$$

It has degree 1168, proving the product degree 1177. This alternate weight is proved here only for positive supplied coordinates. The 26-integer extension in the next section uses the main weight $W = v^2 + D + 2b$.

7 A polynomial in 26 unrestricted integers

7.1 A larger admissible signed domain

The same equations are sound with only the following seven coordinates required to be positive:

$$J, F, \alpha, z, s, \eta, \zeta.$$

Allow h, T, r to be arbitrary integers, assume $i \neq 0$, and allow $w \geq 0$. This extension needs two boundary arguments; it is not obtained just by changing the declared domain.

Transport still gives $C \geq 0$ and $R > 0$, so $b > 0$. The conjugate bound remains valid with $|v|$ in place of v:

$$v^2 + D + 2b \geq 2(|v|\sqrt{D} + b) > |y'|.$$

If $w > 0$, the same separation proof recovers rational t, u, y. If $w = 0$, then $E = \ell = 0$, so $A = 1$ and $t = \pm 1$ is already rational. The second stage of separation still proves that u, y are rational.

When $T \neq 0$, recover $f \in \mathbb{Z}$ as before and use Lemma 3.1. When $T = 0$, $v = B_{\text{aux}} = 0$; instead the root relation directly reads

$$(Q-1)y^2 = Qb^2 - 1.$$

The same lemma, with $V = b$ and $m = |\Delta i c^2|$, still proves $y \in \mathbb{Z}$. Thus the main congruence and integral $\gamma = r - Wu - Lt - y$ are available in all cases.

If $w = 0$, then $X = 0$, $a = Y$, and $u = ac + \gamma H$. The main norm would give

$$1 = u^2 - \Delta c^2 = H(2ac\gamma + H\gamma^2 - c^2),$$

impossible because $H > 1$. Therefore $w > 0$.

Now the authentic bounds give $R < q^4$ and $\Delta > a^2 \geq q^6$, hence $\Delta > R$. For $T = 0$, the integer $b = c + RD$ satisfies $1 < b < Q$, because

$$Q - b = \Delta i^2 c^4(\Delta - R) - c - R > 0$$

using $c \geq 2$. But then

$$y^2 = b^2 + \frac{b^2 - 1}{Q - 1}$$

lies strictly between b^2 and $(b+1)^2$, impossible. Thus $T \neq 0$, so f is recovered in every actual solution.

The inherited signed interface now applies with the positive ordinary multiplier $\Delta|i|$, and proves blank halting. The sign of h is harmless: once $E > R + 2$, the relation $k = R + \varepsilon + hE > 0$ forces $h \geq 0$, after which the usual index argument applies.

7.2 Completeness with odd i and square w

The source package 14 proves a canonical witness can be chosen with i odd and w a nonzero square [5]. We recall why the new coefficient recipe preserves its prerequisites. The spatial width and temporal height may both be sufficiently large powers of five. The coefficient exponent $d_{\text{cell}} = 5^{15}$ is odd, so the exponent t_{pow} in $q = 2^{t_{\text{pow}}}$ is odd. The canonical tuned rank is $R \equiv 3 \pmod 4$, hence odd, and

$$w = 2^{R - t_{\text{pow}}}$$

is a square.

For the normalized strong witness, let $c = \psi_{a+2}(R)$, which is odd, and choose the auxiliary index $m = cR$. Then $c^2 \mid \psi_{a+2}(cR)$ by the binomial expansion of the c-th power of the Pell unit at index R. Thus

$$i = \psi_{a+2}(cR)/c^2$$

is a positive odd integer. The same odd-index quotient identities give integral positive auxiliary witnesses, as proved in package 14. The new language permits precisely the padding and five-adic tuning used in that proof. Finally define r by (23).

7.3 The explicit substitution

Use the positive surjection

$$\mathrm{Pos}(x, y, t) = 1 + x^2 + y^2 + t(t + 1).$$

It is positive for all integers and represents every positive integer. For $n \geq 1$, the three-square theorem represents $4n - 3$. Modulo four exactly one square is odd and two are even; writing them as $(2t + 1)^2, (2x)^2, (2y)^2$ gives the formula above.

Replace each of the seven positive ports by its own Pos, and set

$$i = 2j + 1, \qquad w = u_0^2,$$

leaving h, T, r unrestricted. The resulting polynomial $P_{26}^{\mathbb{Z}}$ has

$$7 \cdot 3 + 5 = 26$$

unrestricted integer variables. The soundness and completeness arguments just given show

$$\mathrm{ZFC} \vdash \Big(\forall \mathbf{z} \in \mathbb{Z}^{26}\ P_{26}^{\mathbb{Z}}(\mathbf{z}) \neq 0\Big) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

The seven positive substitutions cost 49 gates; the odd multiplier and square scale cost three more. Hence the product uses 168 operations. The corresponding weighted degrees are

$$\deg N = 2640, \qquad \deg N_k = 13, \qquad \deg N_t = 4,$$

giving exact product degree 2657. The degree proof is uniform in $\beta > 0$, and the delivered audit records a nonzero leading monomial.

8 The smaller complete coefficient tuple

8.1 A stricter finite language

The machine remains the literal 215-state K215, with 430 transitions and a separate halt state. Its dense table has SHA-256

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

Halt is completed by a stationary transition only for padding an already halted computation; the original transitions entering halt are unchanged.

Retain the 439 cell symbols from T4 [3]: headless bits $0, 1$; headed code $2(q + 1) + b$, $0 \leq q \leq 215$, $b \in \{0, 1\}$; initialization symbols $L_0 = 434, Q_0 = 435, R_0 = 436$; vertical boundary $V_0 = 437$; horizontal boundary $H_0 = 438$. A tile is (l, c, r, n), representing left, center, right, and next.

For every tile other than the special Forbidden tile, impose the old T4 predicate and three additional guards:

1. Both pairs (l, c), (c, r) obey the symmetric row rule. Away from a vertical boundary, symbols have the same kind: ordinary, initial phase, or horizontal boundary. Initial-phase pairs are $L_0L_0, L_0Q_0, Q_0R_0, R_0R_0$. A pair containing V_0 is allowed unless both entries are V_0.

2. If V_0 occurs among l, c, r, every ordinary symbol in that triple is blank and headless.

3. If $n = H_0$, every head among l, c, r is in halt state 215.

Retain Start $= (434, 435, 436, 1)$ first and Forbidden $= (438, 0, 0, 438)$ second. Forbidden remains an old-valid local tile, but its selector is prohibited by the mask. Append all other strict tiles in lexicographic order.

There are 5279 strict tiles and 5280 including Forbidden. An exhaustive enumeration and a separate center-class census agree:

Center	Headless	Working head	Halt head	L_0	Q_0	R_0	V_0	H_0
Count	3495	1720	16	6	1	6	24	12

Every new tile is an old tile, so soundness is inherited by taking a subset. For completeness, pad a finite blank halting computation with at least three never-visited blank cells at each vertical wall, and close its last row after halting. The new guards hold throughout, including helical seams. Arbitrarily large independent padding in width and height is still possible, including powers of five.

8.2 All fixed scalars and finite sums

In this subsection use A_{cl} for the clause radix, to avoid confusion with the first Pell radicand A. Set

$$a_{\mathrm{cell}} = 439, \quad k_{\mathrm{tile}} = 5280, \quad A_{\mathrm{cl}} = 8192, \quad \rho = 2^{78125}, \quad B = \rho^{390625} = 2^{5^{15}}.$$

For $0 \le u \le 438$, define

$$e_l(u) = 7036 + u, \quad e_c(u) = 6597 + u, \quad e_r(u) = 6158 + u, \quad e_n(u) = 5280 + u.$$

The support $\mathcal{E}$ is the union of selectors $0, \dots, 5279$, these four copy intervals, dummies $7475, 7476$, and the four ordered anchors

$$7480,\ 22440,\ 14960,\ 44880.$$

There are 7042 support positions. For tile $w_j = (l_j, c_j, r_j, n_j)$, put

$$v_j = 1 + A_{\mathrm{cl}}^{1+l_j} + A_{\mathrm{cl}}^{440+c_j} + A_{\mathrm{cl}}^{879+r_j} + A_{\mathrm{cl}}^{1318+n_j} + \sum_{t=1}^{4} A_{\mathrm{cl}}^{1756+t}.$$

At the four copy intervals use respectively

$$A_{\mathrm{cl}}^{1+u}, \quad A_{\mathrm{cl}}^{440+u}, \quad A_{\mathrm{cl}}^{879+u}, \quad A_{\mathrm{cl}}^{1318+u}.$$

At the ordered anchors use A_{cl}^{1756+t}, $t = 1, \dots, 4$; at the dummies use zero. This specifies v_e at every $e \in \mathcal{E}$. Let

$$\mu = A_{\mathrm{cl}} - 2 + \sum_{j=1}^{2849} A_{\mathrm{cl}}^{j}.$$

The complete constants are

$$\begin{aligned}
D_C &= \rho^{1317} + \rho^{69077} + \rho^{7480} + \rho^{22440} \\
&\quad + \sum_{e\in\mathcal{E}} v_e\big(\rho^{158838-e} + \rho^{248600-e}\big), \\
D_R &= \rho^{68638}, \\
M_F^0 &= \mu(\rho^{158838} + \rho^{248600}) + \sum_{u=0}^{438} \rho^{6597+u} \\
&\quad + \sum_{u=0}^{438} \big(\rho^{68638+7036+u} + \rho^{68638+6597+u}\big) + \rho^{14960} + \rho^{44880}, \\
M_C &= B - 1 - \sum_{\substack{e\in\mathcal{E}\\ e\neq 1}} \rho^e - 2\rho^{7475}, \\
M_F^{\mathrm{nat}} &= M_F^0 + 4.
\end{aligned}$$

Finally,

$$\boxed{(\beta, K_0, M_C, M_F) = (B-1,\ D_C + BD_R,\ M_C,\ M_F^{\mathrm{nat}} + B - 1).} \tag{25}$$

These are four particular integers, defined by bounded finite enumerations and sums. Their definition never asks whether K215 halts. The JSON recipe and literal alphabet fix every ordering choice.

8.3 Why the modified packing still works

The copy intervals are packed more closely than in T4. The new anchor set is $\{M, 2M, 3M, 6M\}$, with $M = 7480$. The anchor tests map M to $2M$ and $3M$ to $6M$, using shifts $M, 3M$. The difference $4M$ has exactly one ordered realization in the whole support, namely $(2M, 6M)$. All nonanchor positions are below M, and the cell circumference is more than twice the support diameter. Therefore a whole-radix rotation preserving both target anchors has zero cell displacement.

The scalar geometry is

$$E_{\max} = 44880, \quad H_c = 68638, \quad T_1 = 158838, \quad T_2 = 248600, \quad L_{\mathrm{cell}} = 390625.$$

The two clause bands have gap $89762 > 2E_{\max}$, congruent to 2 (mod 4), and

$$T_2 + 2E_{\max} + 2 = 338362 < L_{\mathrm{cell}}.$$

Thus at least one clause band is clean under every support rotation. The inherited no-carry and odd-anchor argument first forces a binary rotation to be a radix rotation; the unique anchor difference then forces whole-cell synchronization.

The verifier checks all $1319 = 3 \cdot 439 + 2$ copy and anchor convolution targets by exact finite set identities. It also checks clause-band isolation, the full support population, and the original conservative carry bound. The clause target has population $\mathrm{pc}(\mu) = 2861$, so

$$\mathrm{pc}(M_F^{\mathrm{nat}}) = 7042, \qquad \mathrm{pc}(M_C) = 5^{15} - 7042.$$

The radix exponent 78125 exceeds both the 37053-bit conservative threshold and the direct coefficient-mass threshold.

Both $78125 = 5^7$ and $390625 = 5^8$ are powers of five. Because M is divisible by four and $T_2 - T_1 \equiv 2$ (mod 4), the actual coefficient residue remains

$$2D_C - D_R \equiv 3 \pmod 5.$$

This is checked by summing the actual finite clause coefficients, as well as by the band-cancellation argument. Consequently the original dummy-bit subset adjustment still tunes the actual packed rank to the required congruence modulo $d_{\mathrm{cell}}N_{\mathrm{cell}}$, with $d_{\mathrm{cell}} = 5^{15}$. The second dummy need not be used.

The full adapter proof is in `proofs/REDUCED_T4_PROOF.md`. It identifies which parts are inherited compiler lemmas and which are new finite support identities. This preserves the inputless compiler interface used above.

9 Reproducibility, evidence, and limits

9.1 What is supplied

The archive contains this LaTeX source and its compiled PDF, the nine JSON arithmetic circuits, the two standalone circuit generators, the complete 5280-tile alphabet, the coefficient recipe, the literal K215 transition table and JSON, the finite compiler checker, independent proof/audit notes, selected pinned upstream proofs, source hashes, and a single reproduction entry point. The inherited 67-operation circuit is also included as `upstream/PRODUCT67.json`; the coefficient binding records its port aliases for the new tuple. The arithmetic and finite checks require only Python's standard library.

From the extracted package directory:

```
python3 verify.py
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

The verification driver regenerates and checks the circuits and finite alphabet. It does not expand the final four huge integers.

9.2 Checks and their interpretation

The algebra checks compare the norm circuit with a separately implemented product of all eight conjugates in an eight-dimensional multiquadratic quotient algebra. They include signed off-solution assignments, parity identities, all-port liveness, exact operation counts, integer-substitution identities, and leading-form certificates. The uniform symbolic leading forms, rather than numerical sampling, prove the stated exact degrees.

The coefficient checker binds all 430 machine transitions to the literal table hash, examines all 46980 horizontal triples surviving the one-head filter, and tests all 439 successors for the 13465 triples surviving the row filter. This is $5,911,135$ explicitly tested quadruples. Every skipped triple violates a stated guard. It checks the class count, every convolution target, the support and anchor conditions, mask populations, carry estimates, and actual coefficient residue modulo five. It also tests 1143 helical windows from two deliberately halting control machines.

These checks support concrete finite and algebraic claims. They do not establish the unbounded compiler theorem by testing, provide an accepting K215 computation, or prove ZFC consistent.

9.3 Formalization boundary

The reviewed K215 verification records report the exact dense-machine endpoint in Lean and Rocq [7]. We fetched and inspected the endpoint and its recorded scope; we did not rerun those large builds. No new end-to-end kernel proof of the norm reduction or coefficient adapter is claimed. The delivered proof is an ordinary mathematical proof composed with explicitly named inherited lemmas.

There is no global optimality claim. In particular the repository already discusses an abstract nine-unknown existence theorem derived from classical universal representations [1]. This report supplies a smaller explicit instance in the concrete compiler family; it does not improve the universal variable frontier. It also does not improve the family's 67-operation minimum or its low-degree systems.

Finally, (1) is an equivalence, not a resolution of Con(ZFC). Under the usual incompleteness hypotheses, consistency gives unprovability of the no-solution statement in ZFC. Two-sided undecidability requires an additional soundness hypothesis; we do not infer it from consistency alone.

Sources and attribution

All repository sources below are pinned to `f0ae4e0991ca44a7bb4e5671d42e33d775e48035`. The accompanying `SOURCE_MANIFEST.json` gives exact paths and hashes of included source copies. The upstream ProveIt and machine notices are preserved in the archive.

References

[1] Vladimir Reshetnikov et al., *ProveIt: Con(ZFC) in Diophantine form*, source snapshot above. https://github.com/VladimirReshetnikov/ProveIt. The reconciliation README in the named report directory identifies packages 09–15 and their stated metrics.

[2] Package 09, *Smaller Diophantine descriptions of ZFC consistency*. Files `con_zfc_reductions.tex`, `proofs/ROOT_AMALGAMATION_PROOF.md`, and the accompanying circuits. Weighted three-root amalgamation and weak sign recovery.

[3] Package 11, *Combining explicit Diophantine encodings of ZFC consistency*. Files `bridge/BRIDGE_AUDIT.md`, `bridge/T4_CONSTANTS.json`, and `bridge/check_t4.py`. The original T4 coefficient compiler.

[4] Package 12, *Smaller Diophantine forms of ZFC consistency*. `proofs/ORDINARY_BLANK_PROOF.md` and `proofs/ROOT_ELIMINATION_PROOF.md`. Ordinary strong Pell interface and signed main-root soundness.

[5] Package 14, *Con(ZFC) synthesis*. `proofs/signed_domain_proof.md`, `proofs/independent_signed_review.md`. The seven-positive-port extension, odd multiplier, and square scale.

[6] Retained wave-1 package 06 proofs, bundled in the current packages: `blank_helical_proof.md`, `fixed_blank_67_proof.md`, and `pell_lemma_audit.md`. Blank-input computation tableaux, actual-rank tuning, and Pell lemmas.

[7] ProveIt, `Computability/BusyBeaver/Lean/BusyBeaver/ZFC/KEndpoint215.lean`, and `Computability/BusyBeaver/Research/ZFC/k225-formal/verification/k215/README.md`. Exact-machine calibration and formal verification records.

[8] This package: `proofs/NORM12_PROOF.md`, `proofs/NORM12_INDEPENDENT_AUDIT.md`, `proofs/INTEGER26_PROOF.md`, `proofs/REDUCED_T4_PROOF.md`, and the generated receipts. New mathematical derivations and finite verification evidence.